Systems Analysis and Control

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Lecture 21: Stability Margins and Closing the Loop

Overview

In this Lecture, you will learn:

Closing the Loop

- Effect on Bode Plot
- Effect on Stability

Stability Effects

- Gain Margin
- Phase Margin
- Bandwidth

Estimating Closed-Loop Performance using Open-Loop Data

- Damping Ratio
- Settling Time
- Rise Time

Review

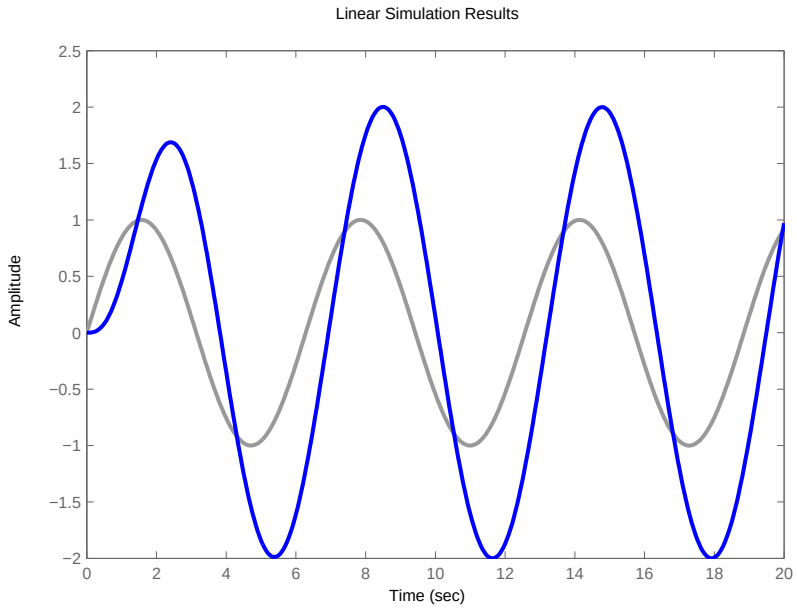
Recall: Frequency Response

Input:

$$u(t) = M\sin(\omega t + \phi)$$

Output: Magnitude and Phase Shift

$$y(t) = |G(\imath\omega)|M\sin(\omega t + \phi + \angle G(\imath\omega))$$



Frequency Response to $\sin\omega t$ is given by $G(\imath\omega)$

Closing The Loop

Now we examine the effect of closing the loop on the Frequency Response.

Use simple Unity Feedback ($K = 1$).

Closed-Loop Transfer Function:

$$G_{cl}(\imath\omega) = \frac{G(\imath\omega)}{1 + G(\imath\omega)}$$

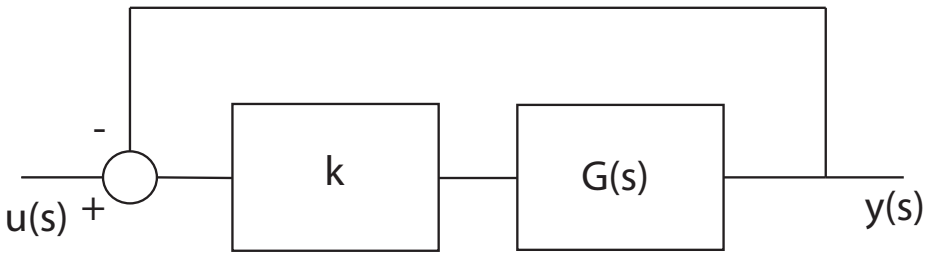


Figure: Unity Feedback

We are most concerned with magnitude:

$$|G_{cl}(\imath\omega)| = \frac{|G(\imath\omega)|}{|1 + G(\imath\omega)|}$$

What Happens to Bode Plot when we Close The Loop?

On the Bode Plot

$$20\log|G_{cl}(\imath\omega)| = 20\log|G(\imath\omega)| - 20\log|1+G(\imath\omega)|$$

Which is the combination of

- The original bode plot
- The new factor $\log|1+G(\imath\omega)|$

We are most concerned with the effect of the new term

$$-20\log|1+G(\imath\omega)|$$

Specifically, as $1+G(\imath\omega)\to 0$

$$\lim_{1+G(\imath\omega)\to 0} -20\log|1+G(\imath\omega)| = \infty$$

Indicates an unstable closed-loop pole!

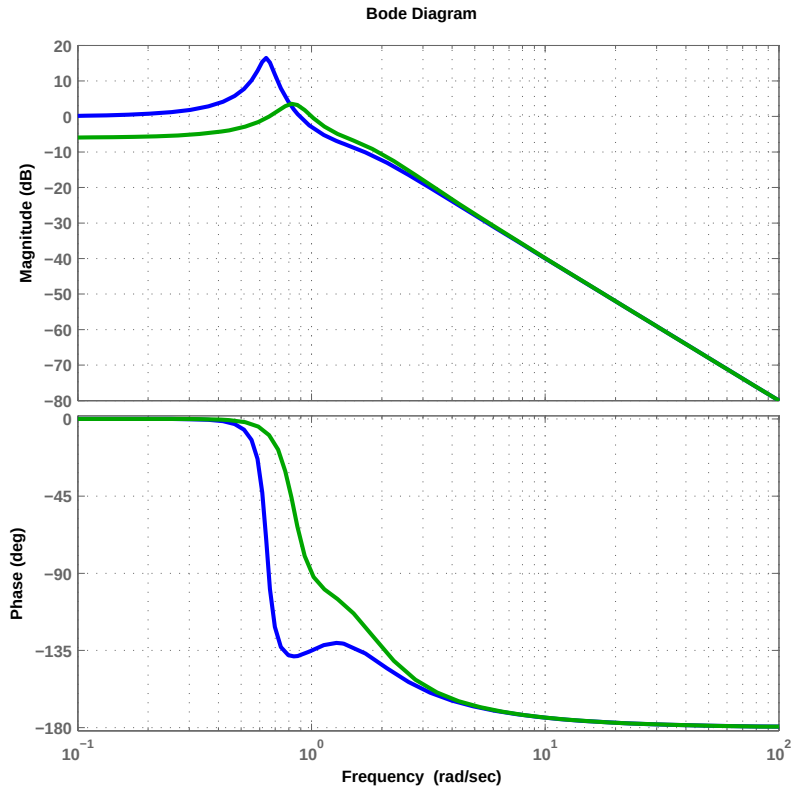


Figure: Open Loop: Blue, CL: Green

Stability Margins: Distance From Closed-Loop Instability

How much can we increase gain/phase before closed-loop destabilizes?

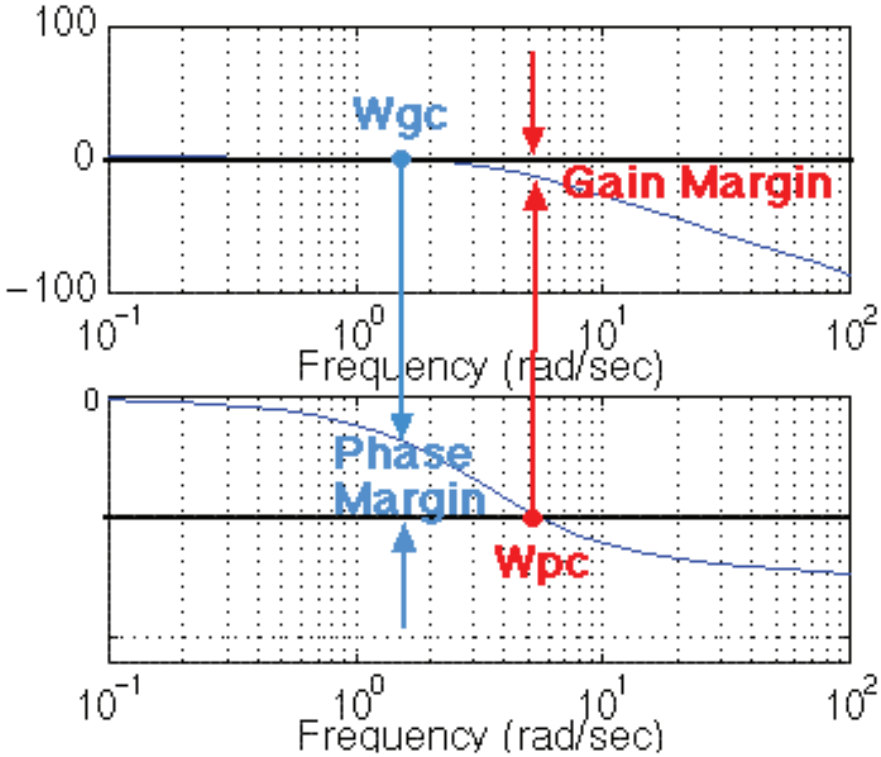
Instability occurs when

$$1 + G(\imath\omega) = 0$$

For this to happen, we need:

- $|G(\imath\omega)| = 1$
- $\angle G(\imath\omega) = -180°$

Stability Margins measure how far we are from the point $(|G| = 1, \angle G = -180°)$.



Definition 1.

The **Gain Crossover Frequency**, ω_{gc} is the frequency at which $|G(\imath\omega_c)| = 1$.

This is the danger point:

- If $\angle G(\imath\omega_c) = 180°$, we are unstable in closed-loop

Stability Margins: Phase Margin

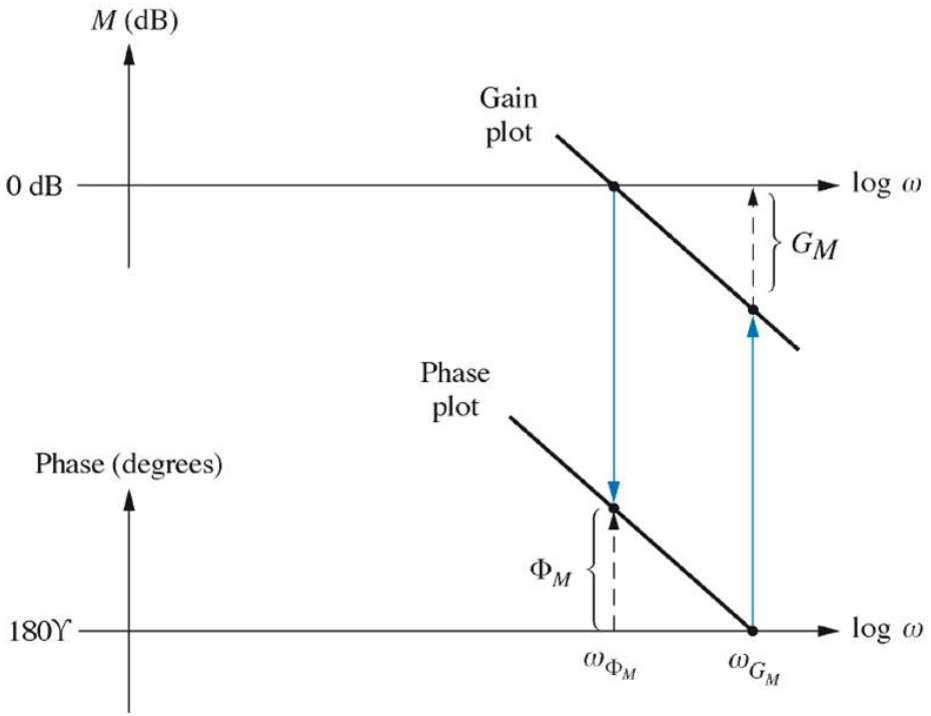
How Much Extra Phase do we have at the **Gain Crossover Frequency**?

Definition 2.

The **Phase Margin**, Φ_M is the phase relative to 180° when $|G| = 1$.

- $\Phi_M = |180^\circ - \angle G(i\omega_{gc})|$

- ω_{gc} is also known as the phase-margin frequency, ω_{Φ_M}



Stability Margins: Gain Margin

How Much Extra Gain is allowed at the **Phase Crossover Frequency**?

Definition 3.

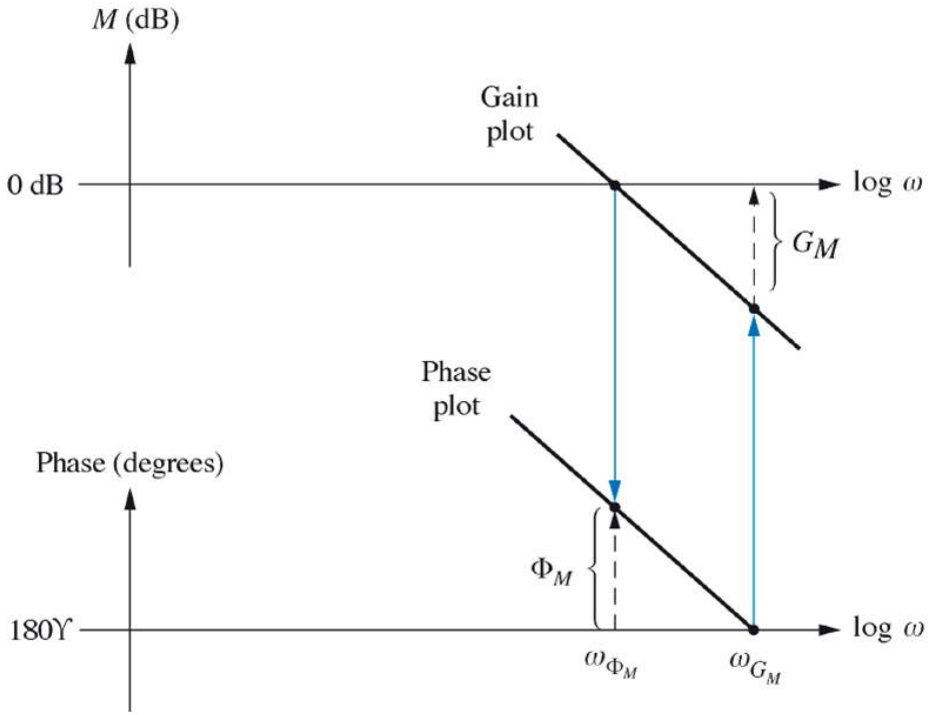
The **Phase Crossover Frequency**, ω_{pc} is the frequency (frequencies) at which $\angle G(\imath\omega_{pc}) = 180°$.

Definition 4.

The **Gain Margin**, G_M is the gain relative to $0dB$ when $\angle G = 180°$.

- $G_M = -20 \log |G(\imath\omega_{pc})|$

G_M is the gain (in dB) which will destabilize the system in closed loop.

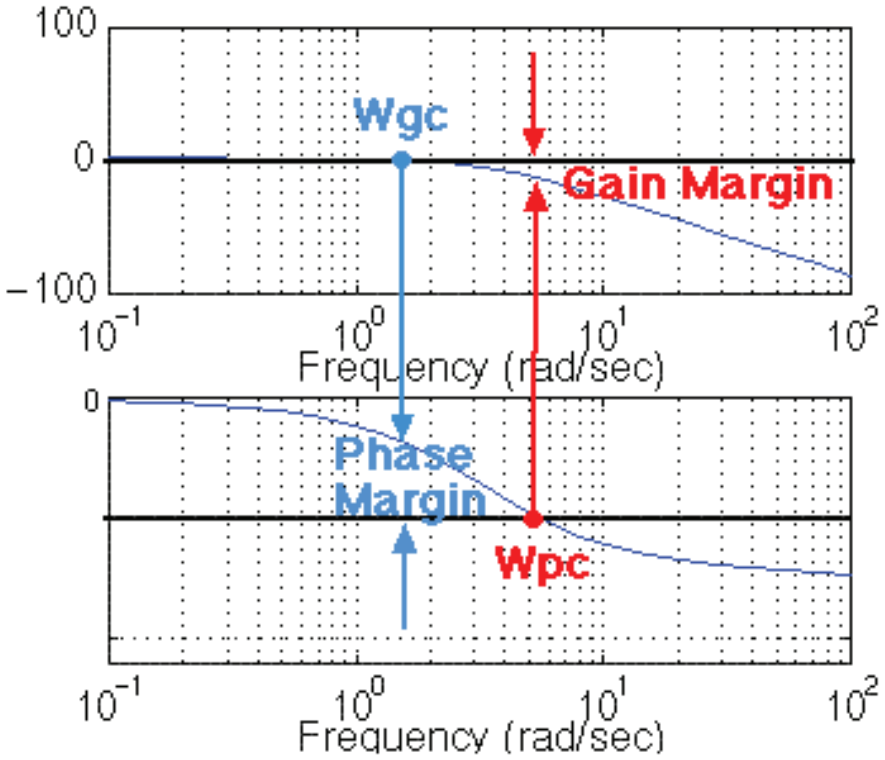


ω_{pc} is also known as the gain-margin frequency, ω_{G_M}

Stability Margins: Gain and Phase Margin

How much extra gain or delay will the closed-loop tolerate?

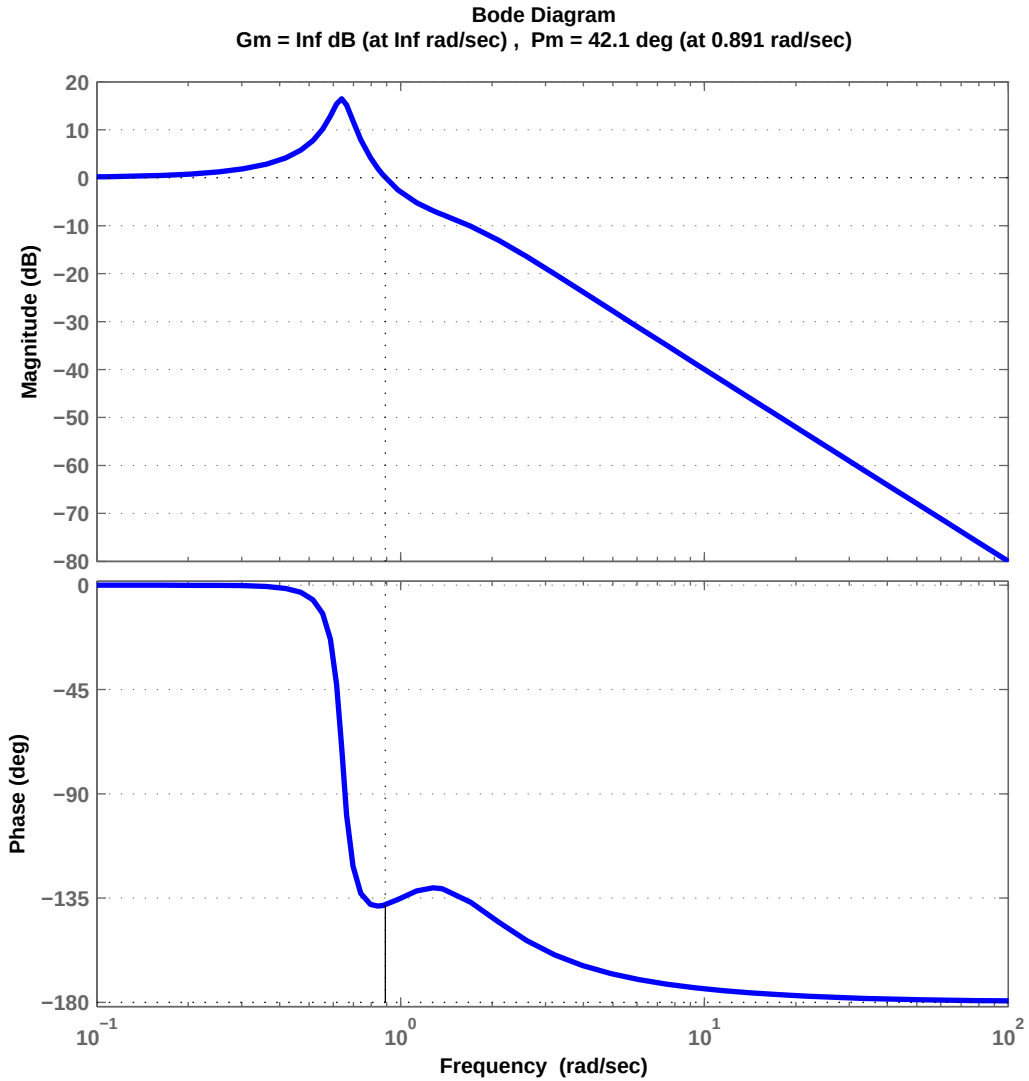
Gain and Phase Margin tell how stable the system would be in Closed Loop.



These quantities can be read from the **Open-Loop Data**.

Illustration of Gain and Phase Margin

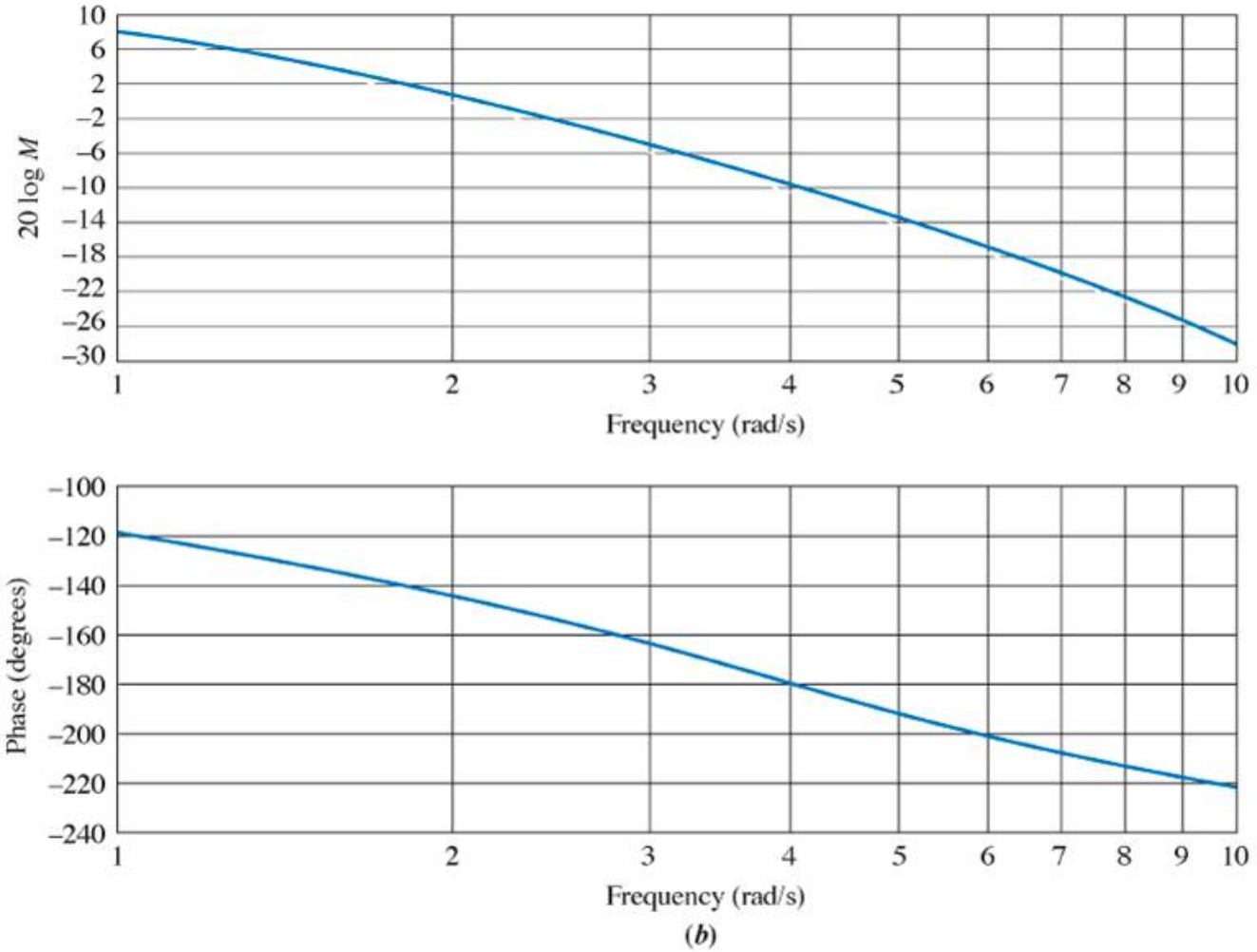
Remember the Suspension System Example?



In Matlab: Use `margin(sys)`

Illustration of Gain and Phase Margin

Draw the Bode Plot by Hand and Read them Off!



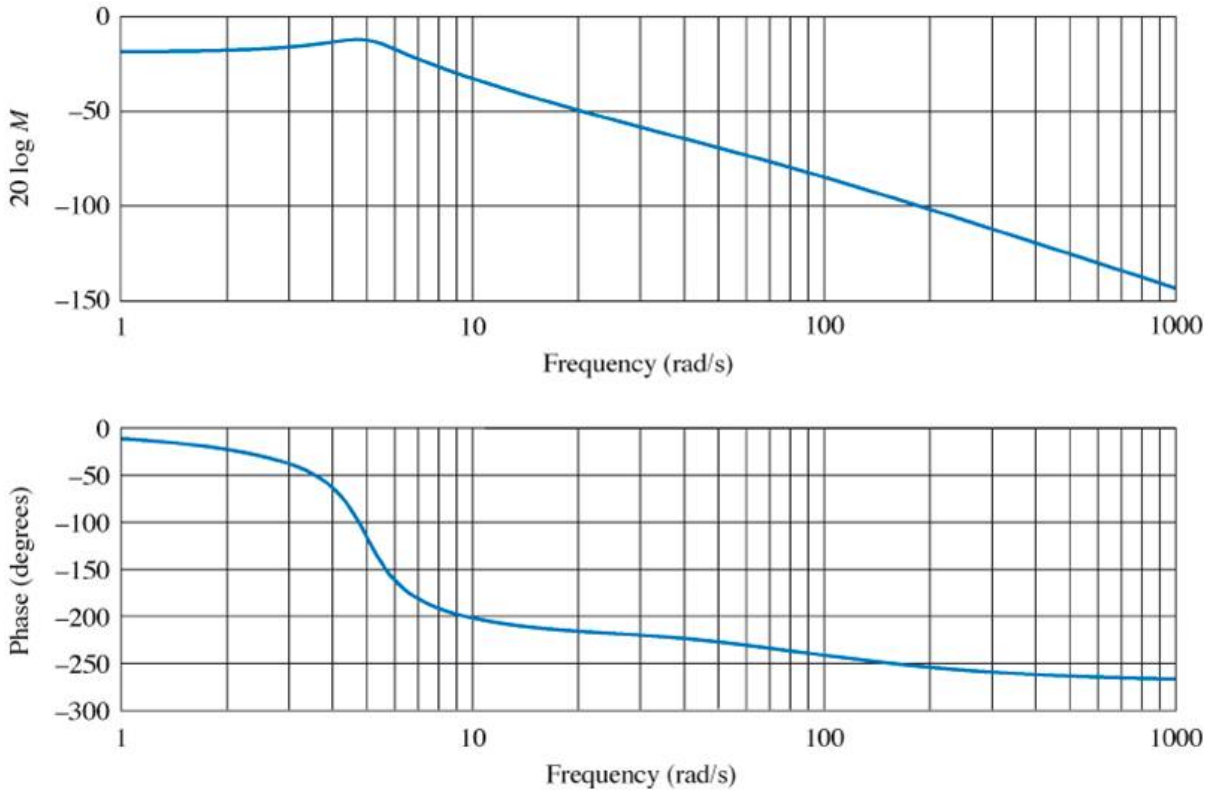
- $\Phi_M = 35^\circ$
- $G_M = 10\text{dB}$

Sometimes Gain and Phase Margin are Undefined

This is GOOD news!

Note that sometimes the margins are undefined

- When there is no crossover at 0dB
- When there is no crossover at 180°



Closed-Loop probably won't destabilize.

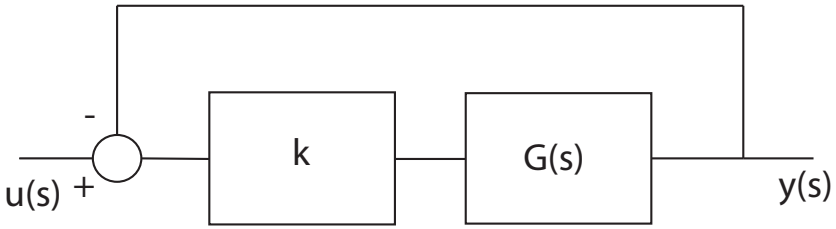
Estimating Transient Response from Bode Plots

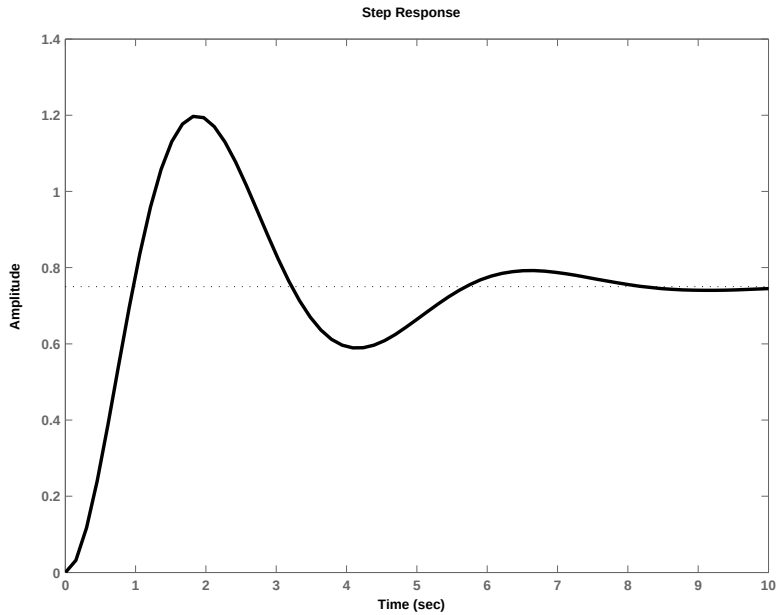
What Happens When We Close the Loop?

Question: How does Closing the Loop Affect Transient Performance?

- We want **Performance Specs!** (T_s, T_r, T_p, M_p)
- We only have open-loop data.
- Φ_M and G_M can help us.

Unity Feedback:





We want:

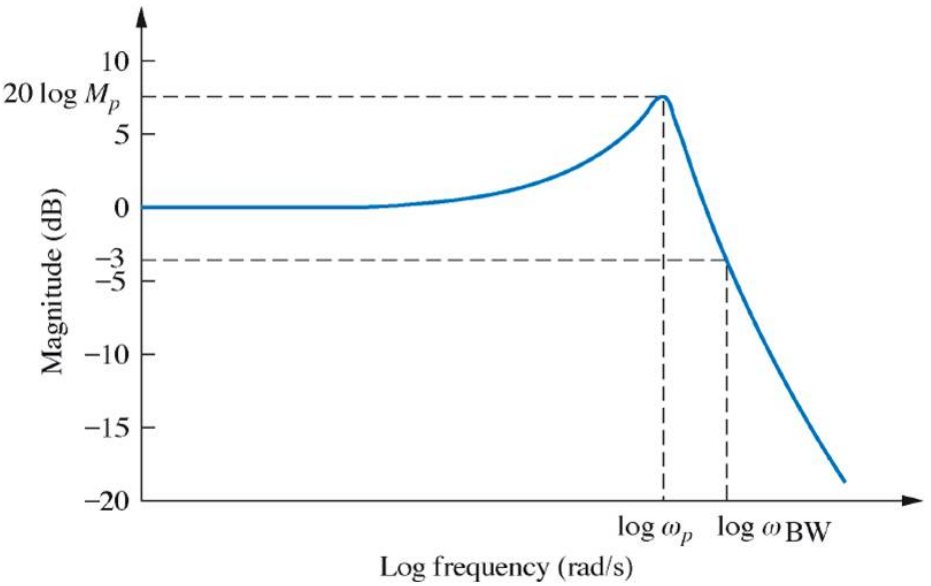
- Damping Ratio
- Settling Time
- Peak Time

New Frequency Response Parameter: Bandwidth

Relation Between Bandwidth (ω_{BW}) and Natural Frequency

Definition 5.

The **Bandwidth**, ω_{BW} is the frequency at gain
$20\log|G(\imath\omega_{BW}) = |20\log|G(0)| - 3\text{dB}$.



- Closely related to crossover frequency.
- The Bandwidth measures the range of frequencies in the output.
- For 2nd order system, Bandwidth is related to natural frequency by

$$\omega_{BW} = \omega_n\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

Problem: We don't know ζ or ω_n!

Closed-Loop Transient Response from Open-Loop Bode

Quadratic Approximation

Assume the closed loop system is the quadratic

$$G_{cl} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Then the open-loop system must be

$$G_{ol} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s}$$

Assume that our open-loop system is G_{ol}

- Use Φ_M from the open-loop to solve for ζ in closed loop!
- We can find an analytic expression for open-loop phase margin (Φ_M) in terms of ζ.

CL Damping Ratio from OL Phase Margin

An approximate linear expression for CL Damping Ratio from OL Phase Margin

Use Open-Loop Phase Margin the **Closed-Loop Damping Ratio** as

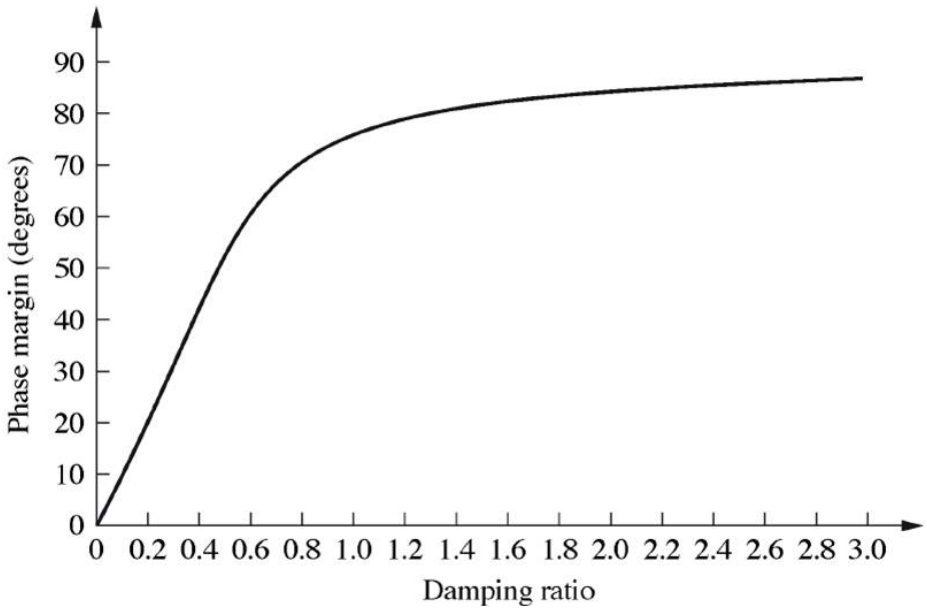
$$\Phi_M = \tan^{-1} \frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{4\zeta^4 + 1}}}$$

Φ_M is from the *Open-Loop Data!*

Approximately

$$\zeta = \frac{\Phi_M}{100^\circ}$$

- Only valid for $\Phi_M < 70^\circ$.
- Given OL Φ_M, we find CL ζ.



Finding Closed-Loop Bandwidth from Open-Loop Data

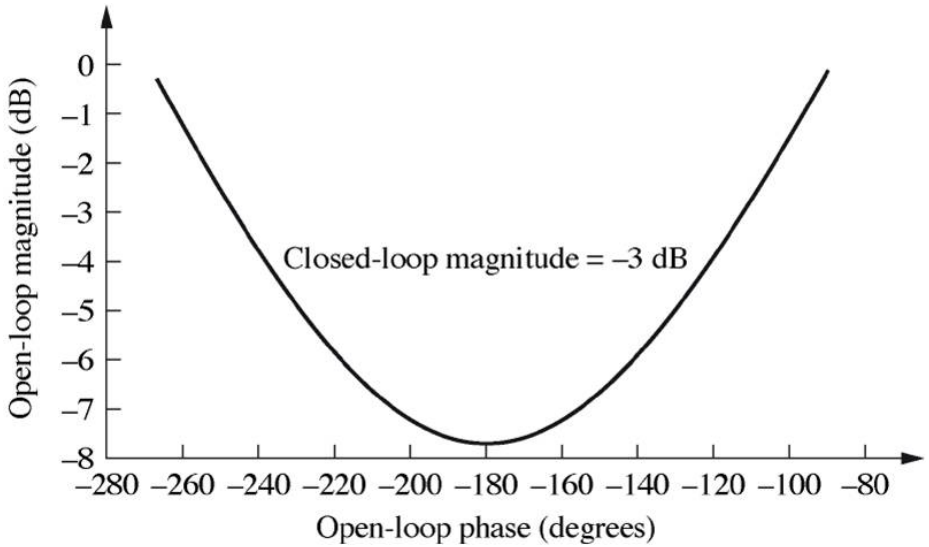
Use the Plot to Match Frequencies (Heuristic)

We found closed-loop ζ from Phase margin.

- Can we find closed-loop Natural Frequency ω_n?
- Given ζ, we can use closed-loop ω_{BW} to find ω_n
- But can we find the closed-loop Bandwidth?

Question: How to find closed-loop bandwidth from open-loop data?

Finding the closed-loop bandwidth from open-loop data is tricky.



Find the frequency on the open-loop Bode plot which intersects this curve!

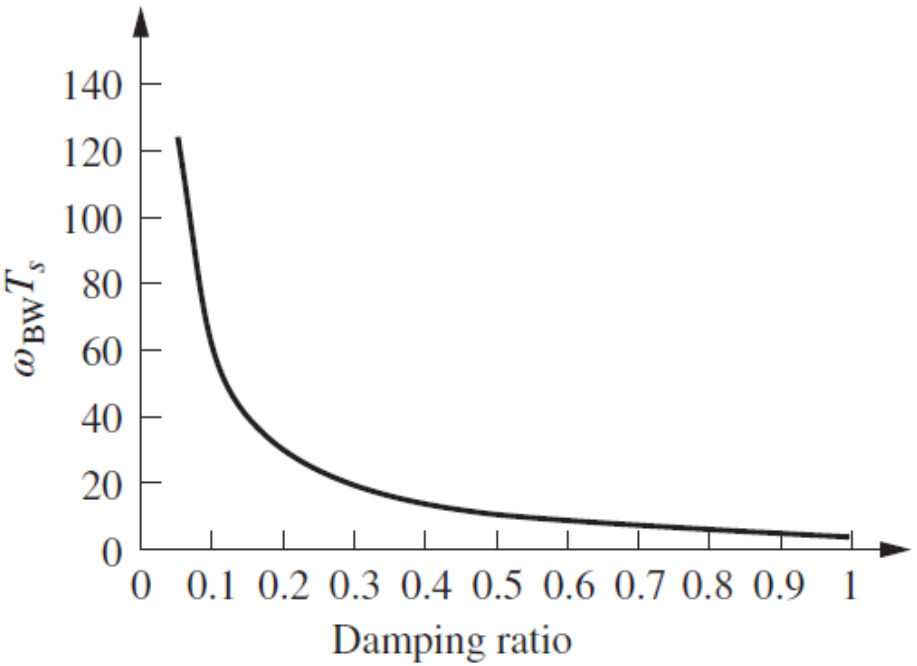
- **Heuristic:** Check the frequency at -6dB and see if phase is $\cong 180°$.

Closed-Loop Settling Time From Closed-Loop Bandwidth

Using our CL ζ and ω_n approximations

We can use the expression $T_s = \frac{4}{\zeta \omega_n}$ to get

$$\omega_{BW} = \frac{4}{T_s \zeta} \sqrt{(1 - 2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$



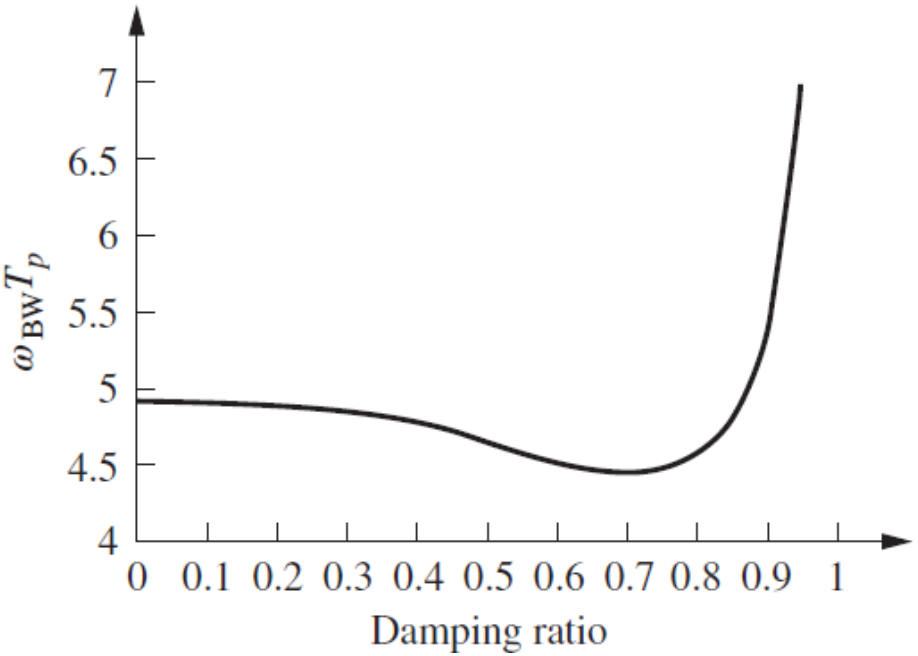
Given closed-loop ζ and ω_{BW}, we can find T_s.

Closed-Loop Peak Time From Closed-Loop Bandwidth

Using our CL ζ and ω_n approximations

We can use the expression $T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$ to get

$$\omega_{BW} = \frac{\pi}{T_p \sqrt{1-\zeta^2}} \sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

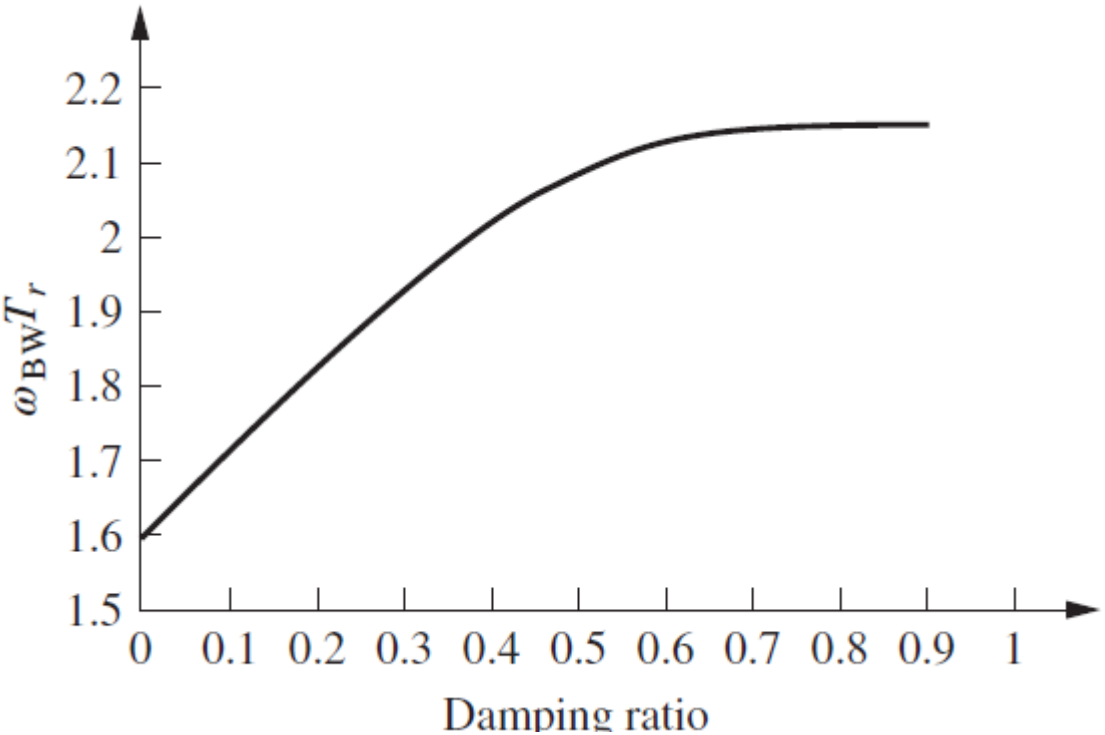


Given closed-loop ζ and ω_{BW}, we can find T_p.

Closed-Loop Rise Time From Closed-Loop Bandwidth

Using our CL ζ and ω_n approximations

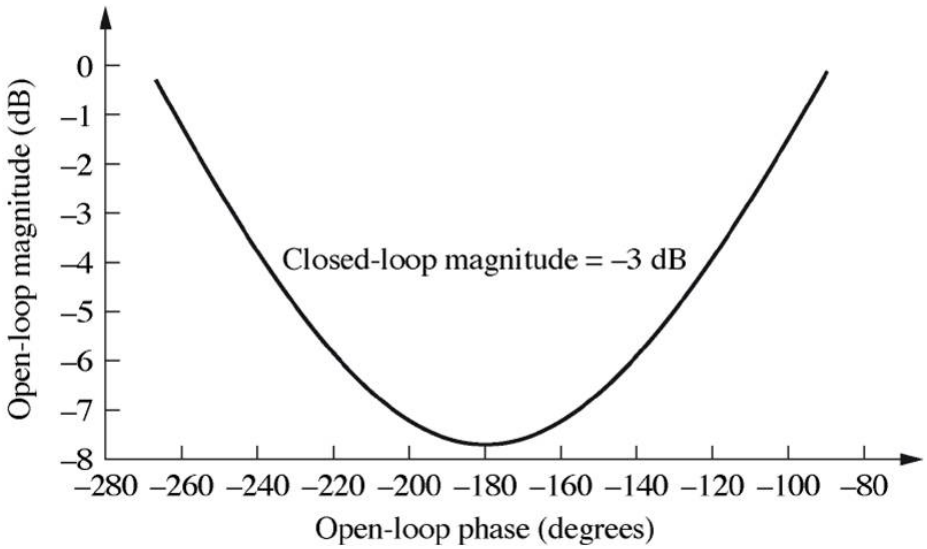
Using an expression for T_r, we get a relationship between $\omega_{BW} \cdot T_r$ and ζ.

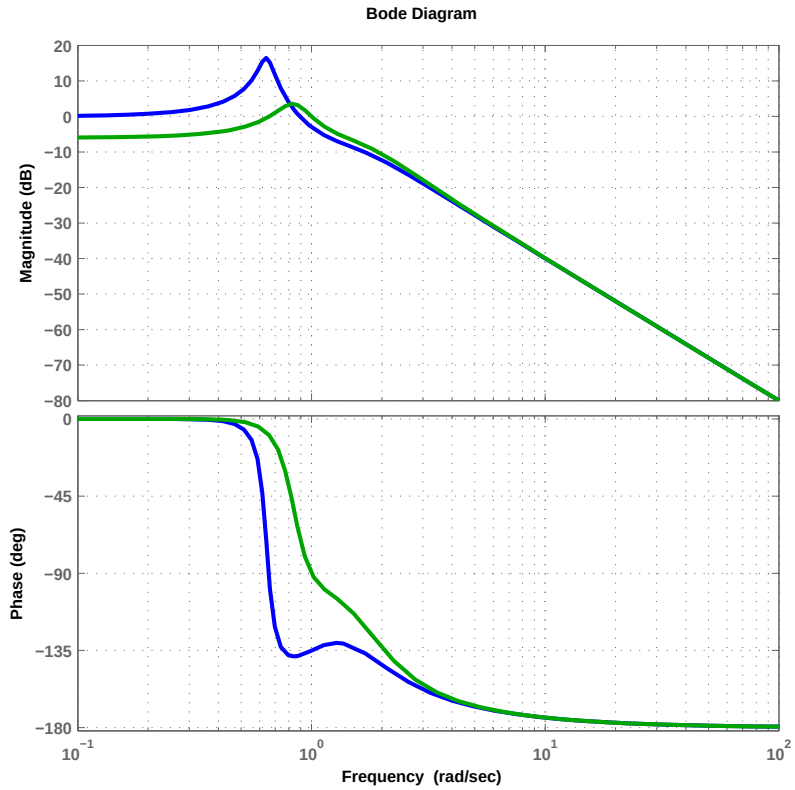


Given closed-loop ζ and ω_{BW}, we can find T_r.

Finding Closed-Loop Bandwidth from Open-Loop Data

Example



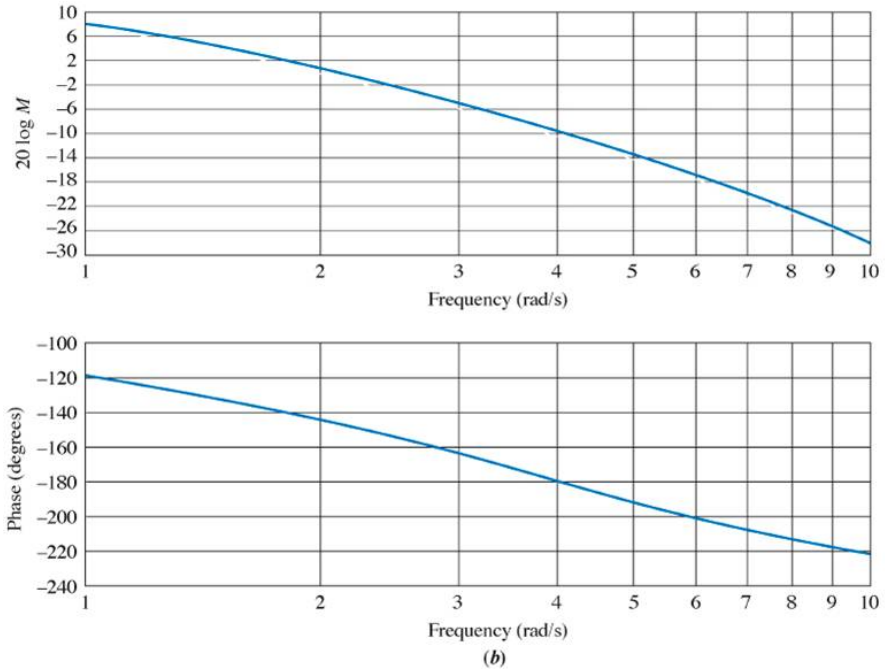


At phase $135°$, -5dB, we get closed loop $\omega_{BW} \cong 1$.

Example of Finding CL Transient Response From OL Data

Step 1: Calculate Open-Loop Phase Margin

Question: Using Frequency Response Data, find T_r, T_s, T_p after unity feedback.



First Step: Find the phase Margin.

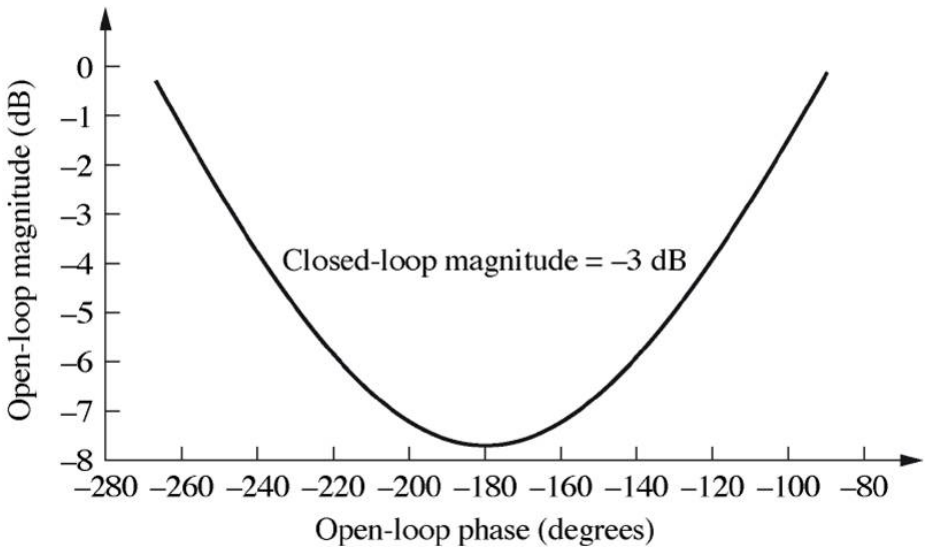
- Frequency at 0dB is $\omega_{gc} \cong 2$
- $\angle G(2) \cong -145^\circ$
- $\Phi_M = 180^\circ - 145^\circ = 35^\circ$

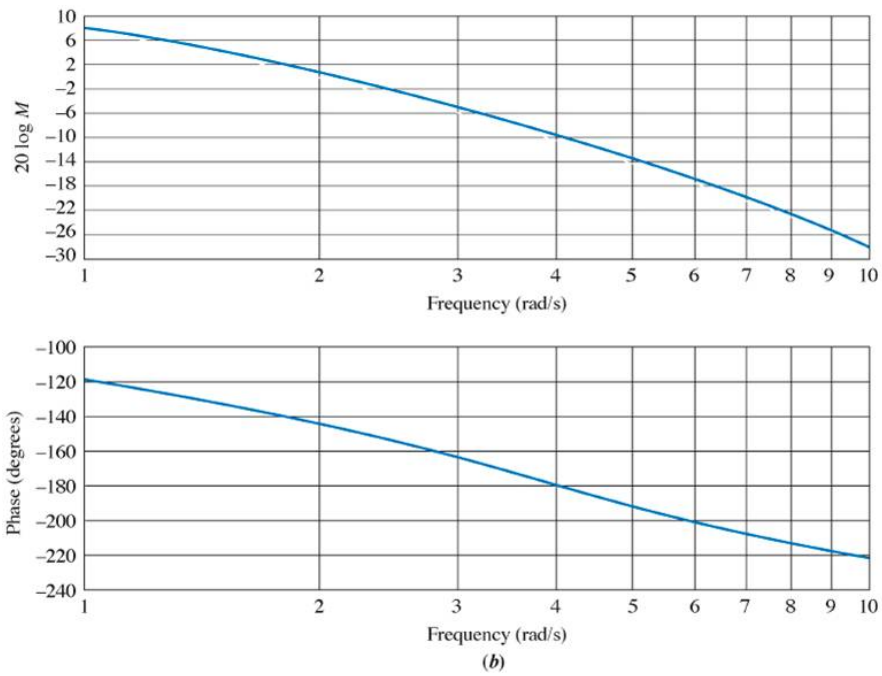
Example of Finding CL Transient Response From OL Data

Step 2,3: Calculate Closed-Loop ζ and ω_{BW}

Step 2: Closed-Loop Damping Ratio

$$\zeta \cong \frac{\Phi_M}{100} = .35$$





Step 3: Closed-Loop Bandwidth

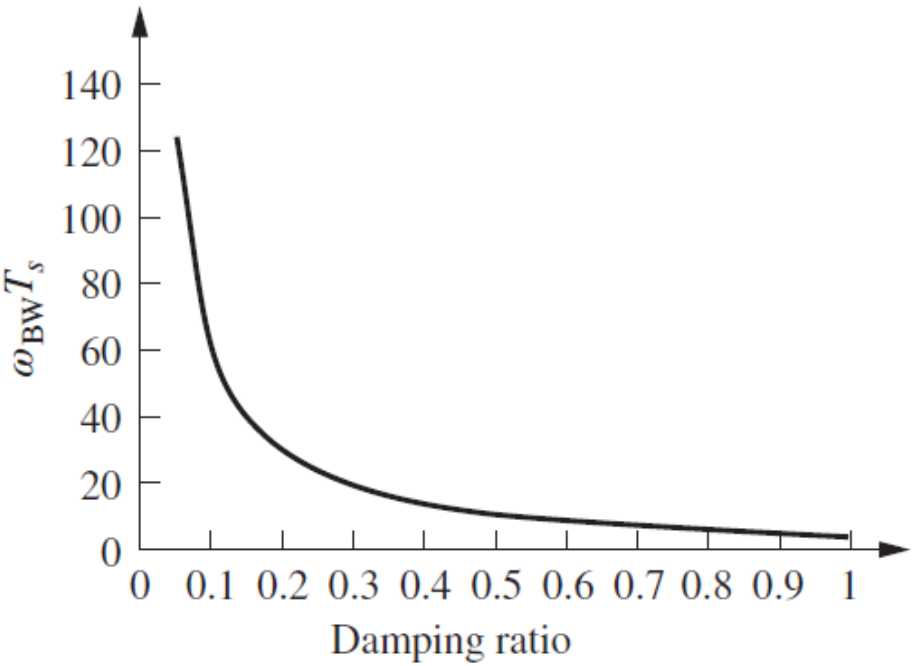
- Intersect at $\cong (|G| = -6dB, \angle G = -170°)$
- Frequency at intersection is $\omega_{BW} \cong 3.7$

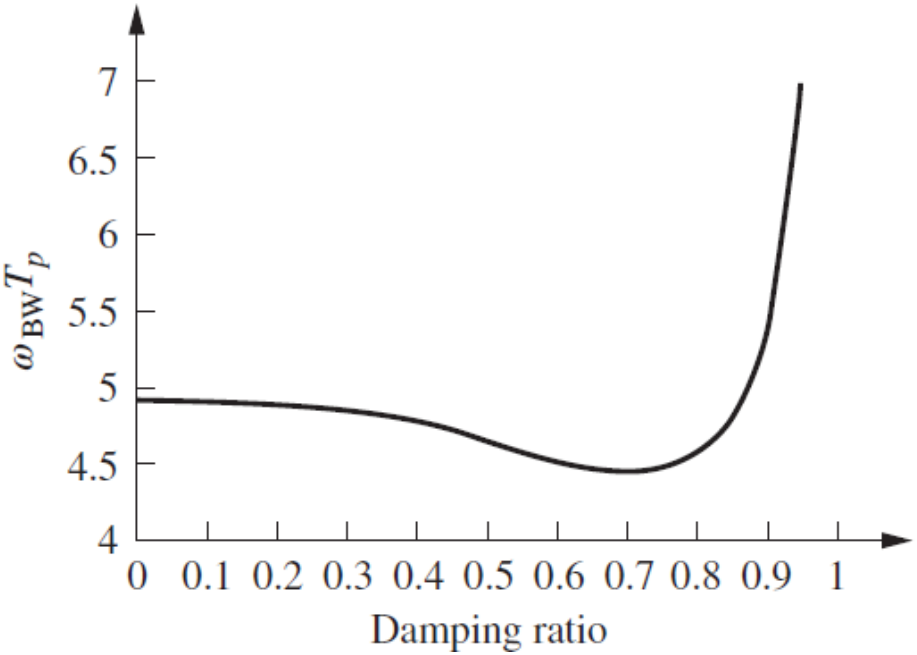
Example of Finding CL Transient Response From OL Data

Step 4: Calculate Closed-Loop Performance Specs

Step 4: Settling Time and Peak Time

- $\omega_{BW} = 3.7$, $\zeta = .35$





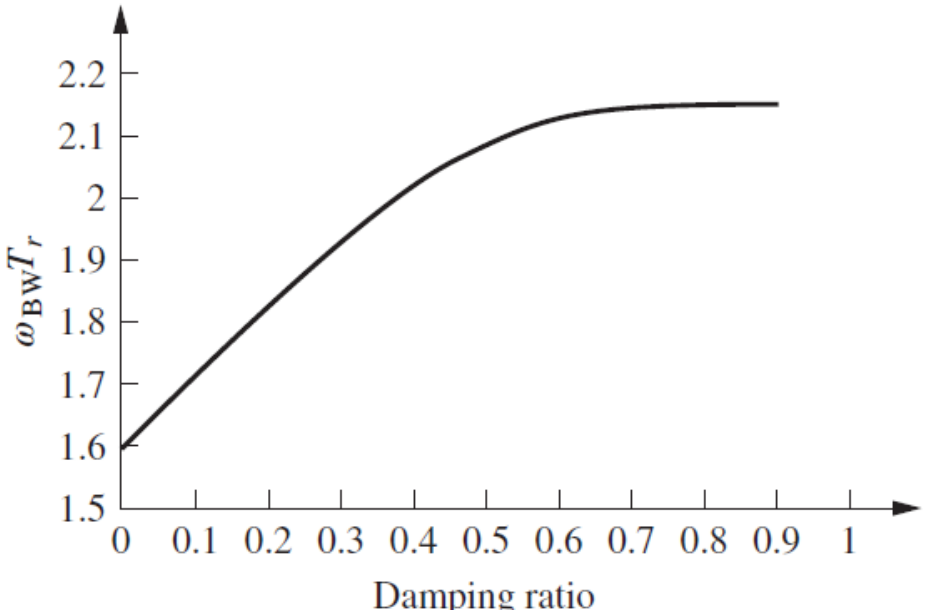
- $\omega_{BW}T_s = 20$ implies $T_s = 5.4s$
- $\omega_{BW}T_p = 4.9$ implies $T_p = 1.32$

Example of Finding CL Transient Response From OL Data

Step 5: Calculate Closed-Loop Rise Time

Step 5: Rise Time

- $\omega_{BW} = 3.7$, $\zeta = .35$



- $\omega_{BW}T_r = 1.98$ implies $T_r = .535$

Example of Finding CL Transient Response From OL Data

Step 6: Experimental Validation

Step 6: Experimental Validation.

Use the plant

$$G(s) = \frac{50}{s(s+3)(s+6)}$$

We find

- $T_p = 1.6s$
  - predicted 1.32
- $T_r = .7s$
  - predicted $.535$
- $T_s = 4s$
  - Predicted 5.4

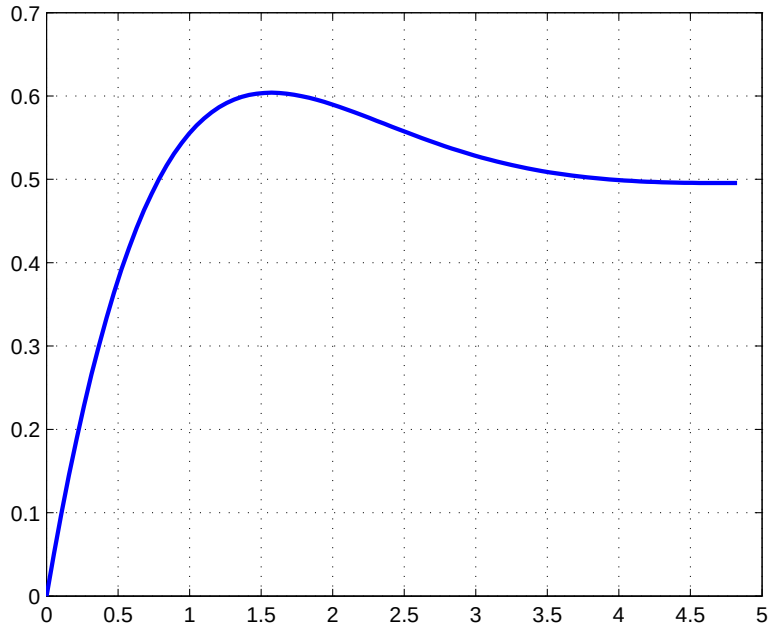


Figure: Step Response

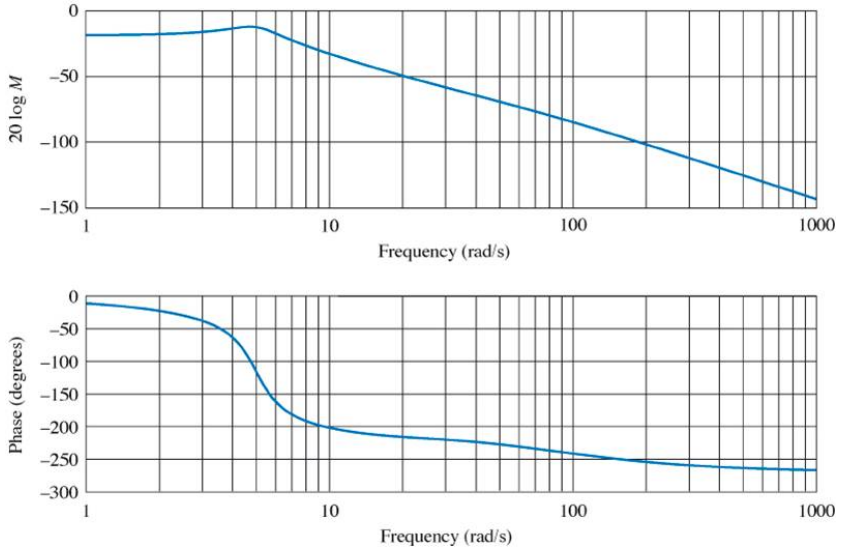
Calculating OL Steady-State Error from Bode

Finally, we want steady-state error.

- Steady-State Step response is

$$\lim_{s\to 0} G(s) = \lim_{\omega\to 0} G(\imath\omega)$$

Steady-state response is the Low-Frequency Gain, $|G(0)|$.



Close The Loop to get steady-state error

$$e_{ss} = \frac{1}{1+|G(0)|}$$

Illustration of Steady-State Error from Bode

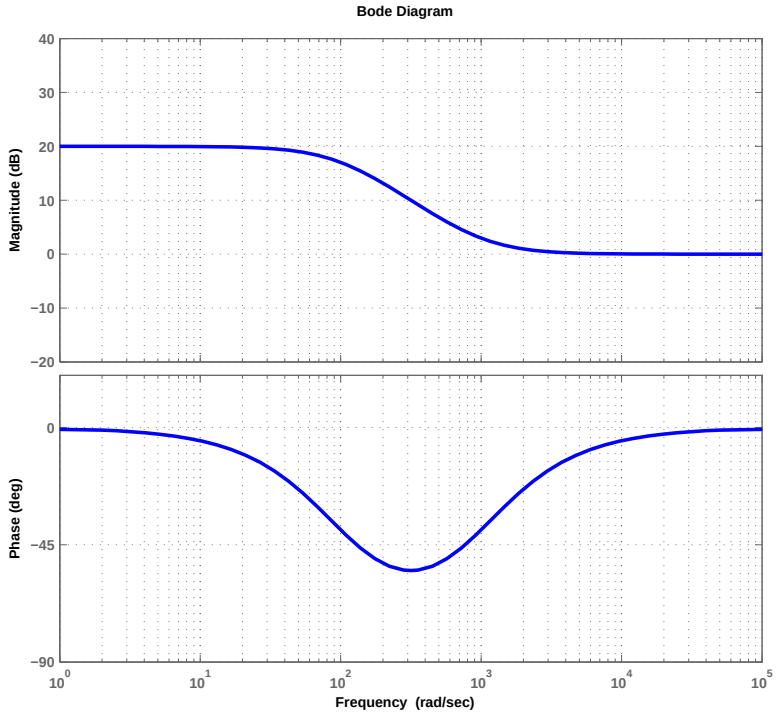
A Lag Compensator

$$\lim_{\omega \to 0} 20 \log G(\imath\omega) = 20dB$$

So $\lim_{\omega \to 0} |G(\imath\omega)| = 10.$

Steady-State Error:

$$e_{ss} = \frac{1}{1 + G(0)} = \frac{1}{11} = .091$$



Summary

What have we learned today? **Closing the Loop**

- Effect on Bode Plot
- Effect on Stability

Stability Effects

- Gain Margin
- Phase Margin
- Bandwidth

Estimating Closed-Loop Performance using Open-Loop Data

- Damping Ratio
- Settling Time
- Rise Time

Next Lecture: Compensation in the Frequency Domain